

USER HANDBOOK | RC3 | 2026-10-08

# From image to a reliable colour workflow

InkProf - When colours have to be right

## The image should reach the paper

For the photographer, the aim is for the carefully crafted image to come into its own in print. A clear, well-controlled workflow helps the photographer, printing business and client understand the result and work towards the same goal.

## Less time and material wasted

For a printing business, documented settings and regular checks provide a stronger basis for consistent deliveries. Finding and understanding errors early reduces the need to make new prints and the extra working time, paper and ink they consume.

## Transparency makes colour understandable

InkProf brings colour targets, spectral measurements, ICC profiles and decisions together in one project. Open source and a traceable workflow reveal how results were produced. This makes it easier to understand difficult colours and work right up to the limits of the chosen printer, paper and ink.

## How this handbook helps

This handbook describes the main window, dialogs and 19 profiling steps, with current profiling paths, smoothing and FWA limitations. Updated on 8 October 2026 against the source. The final appendix follows a complete profiling job and its decisions.

## The three parts of the work

First, document and measure the printer. Next, build and check a profile. Finally, verify a print or document an explicitly limited numerical decision. If needed, another iteration adds more measurements.

InkProf saves TIFF16 for separate printing. Measurement, profile building, analysis and documentation are part of the app workflow. Print using your usual printing application.

## OVERVIEW

# Find your way through the workflow

The row number on the left and the internal step identifier are different things. This handbook uses row numbers to help you find the step in the app.

| Row   | Step in the app                                                      | Purpose / further reading                                       |
|-------|----------------------------------------------------------------------|-----------------------------------------------------------------|
| 1-2   | Create / import RGB target; Save TIFF16                              | Base target and print file. Page 6.                             |
| 3-4   | Measure / select measurement revision; Review and accept measurement | Measure, choose a saved revision and review it. Pages 7-8.      |
| 5-7   | B1. Lock profiling inputs; B2. Create recipe; B3. Build ICC          | Lock inputs, choose a recipe and build a profile. Page 9.       |
| 8     | C1. Fit, Grid and C1                                                 | Numerical checks. Page 10.                                      |
| 9-10  | C2. Save verification target; Measure / select verification revision | Print and measure a verification target. Page 11.               |
| 11-12 | C3. Compare model and print; Review feedback                         | Assess the actual print and feedback. Page 11.                  |
| 13-14 | Approve for intended use; Save ICC and measurement certificate       | Decision and delivery after verification. Page 12.              |
| 15-17 | Save refinement TIFF16; Measure refinement; Build next iteration     | New colour patches and a new profile. Pages 13-14.              |
| 18    | Compare with previous iteration                                      | Compare profiles on a common basis. Page 15.                    |
| 19    | Save ICC + measurement certificate without print verification        | A limited decision without another verification print. Page 16. |

## Reading the status

Ready means the step can be run. Complete means its recorded requirements have been met. Locked means a required input is missing. Stale means an input has changed and the result needs updating. Simply opening and closing a dialog does not complete a step.

## There is more than one route

B2 is needed for the manual profiling route. The profile comparison on row 18 supports your decision. Row 19 is a separate option when the current iteration has been checked numerically but has not been verified by a separate print and measurement.

## Additional guidance

The later chapters cover smoothing, gradients, remeasurement, target preparation, diagnostics and FWA. The final appendix gives the complete recommended profiling route and decision points.

## THE APP HUB

# The main window

First select a row in the table. Read the description and saved results before choosing Run selected step.

**Project tools** | name, settings, log, history and report

**Left: steps 1-19**

Select a row.  
Read the status.

**Right: selected step**

Description and saved results.  
Run the step or open its result.

Schematic guide, not a screenshot. The app also displays status and activity.

## Projects and resuming work

New project creates a project. Open project opens an existing one, including after the app has been closed while prints dry. Project details collects user, material and settings information. The project name and current path appear in the main window.

## Running steps and viewing results

Run selected step opens the next operation or dialog. Open selected step results shows what has already been created. Copy TIFF16 for printing... copies a saved target file for separate printing. Read the result path to make sure you use the correct target and iteration.

## History and checks

Iteration history shows the iterations. Open results log opens the work log. Verify project checks the registered files and project integrity. Refresh reloads the current state; it does not replace Save, Accept or explicit approval in a dialog.

## Reports, views and closing

Open report opens an available report; the label may refer to the certificate. View 3D shows available colour points. View gamut shows a separate ICC-derived 3D surface. Close app provides an explicit way to exit. First finish any active dialog using its intended save or cancel button.

## NEW PROJECT / OPEN PROJECT

# The project remembers your work

A project is a separate folder that keeps settings, files, measurements and history together.

## Create and enter details

Choose New project and enter the name, user and project information. Project details has four tabs: Project and materials, Printing settings, Profiling and Target paper. Use Save to save. Cancel leaves the dialog without saving changes.

## Project and materials

Enter Project name, User, Printer, Paper name, Paper finish, Ink / ink set and Ink type (dye/pigment/mixed/unknown). Use product names precise enough to reproduce the combination. Paper finish includes Glossy and Matte. This tab also contains the FWA / OBA optical-brightener option, explained on the next page.

## Printing settings

Fill in Driver media setting, Printing application, Driver / RIP, Print quality, Colour management, Drying time (hours), Printer coating and Coating settings. Use Printer settings to describe additional choices. The workflow reads the drying time here; recording a duration does not itself establish that the sheet has dried.

## Correct details or start again?

Correct typing errors in Project details. The recipe reads shared information here, and shared project information is edited here. A later FWA choice also updates the project. If the printer, paper, ink or printing conditions will actually form a different combination, a new project is clearer. Follow the app requirements to repeat steps when inputs change.

## Rename and move

Changing the project name in the app also renames its folder. If the folder is renamed outside the app, the app can detect the difference and ask you to confirm the intended name. Move the entire project folder to another computer and open it there. Do not copy only workflow.json or a single ICC file if you want to preserve the complete workflow.

## What is saved?

Project JSON files track identity, settings, steps and iterations. The results log is available as readable text and JSONL, with one log entry per line. Raw data and generated results are linked to each operation. Checksums help the app detect changed inputs.

Pausing while prints dry: finish saving the current step, close the app and reopen the same project when the sheets are ready. Do not create a new project to continue the same measurement session.

## PROJECT DETAILS

# Profile and measurement conditions

## Profiling

Profile name and Description describe the profile. Colour data can be Spectra (D50 / 2 degrees) or Stored XYZ. Spectral data allow recalculation from measured spectra. Stored XYZ preserves the stored colour values. Inverse table (B2A) offers High (denser) or Medium (baseline). A denser table is a calculation setting, not proof of better printing.

## D50 and D65

D50 is the reference illuminant for spectral profiling in the app, together with the two-degree standard observer. D65 is another daylight reference. Assess colour under documented viewing conditions and do not mix comparisons using different illuminants. The current Profiling tab has no separate D65 option; Spectra explicitly specifies D50.

## Optical brighteners: FWA / OBA

FWA (Fluorescent Whitening Agents) and OBA (Optical Brightening Agents) are two names for optical brighteners in paper. OBC (Optical Brightener Compensation) means compensation for their effect: the same phenomenon, but a process rather than a substance. Argyll calls it FWA compensation; X-Rite uses OBC. The algorithms are not necessarily identical. Compensate optical brighteners (D50) selects InkProf's compensation. The supported M0 route models D50 and does not become a physical M1 measurement.

## Change FWA after measurement

Changing only FWA/OBA preserves locked B1 but makes B2 and later profile results stale. Raw measurements remain preserved. Actual changes to materials or printing conditions also invalidate B1. New recipes support FWA before Argyll pre-smoothing and final smoothing; see the M0 white-reference section. See page 21.

## Black point compensation: BPC

Black point compensation maps the black point between colour spaces. It is off in the controlled comparisons described here so that the calculation conditions remain explicit. BPC off describes a condition; it is not an error message.

## Computer and instrument

MATLAB provides the app interface. Python, ArgyllCMS and LittleCMS are used for underlying calculations and tools. Tool paths depend on the computer and must work on the current machine. Check the installation even when the project folder is complete. The program has been tested on macOS. Windows and Linux are not verified environments for the complete InkProf workflow.

## ROWS 1-2 / PAPER DIALOG

# Base target and TIFF16

A base target defines the RGB values the printer should reproduce. The printed and measured version becomes the input for profiling.

### Row 1: create or import

Run Create / import RGB target. Create a target using the dialog options or import compatible existing RGB data. Patch count and identity must remain consistent through layout, printing and measurement. The visual appearance of a target is not enough to identify its measurement definition.

### Row 2: save the print file

Run Save TIFF16 and choose a layout. The app saves a 16-bit TIFF with matching target and layout information. Check page count, patch count and the recorded file path in the result. TIFF16 is the print deliverable; InkProf does not send the job to the printer.

### Target paper and the measurement sled

In Project details, Maximum measurement sweep (mm) sets the greatest width of a sweep. Maximum target length (mm) limits the other target dimension. Roll width (mm) sets the roll width. Limits include margins and are saved in project JSON. Adapt them to your equipment; they are not universal instrument dimensions.

### Editable paper suggestions

Paper planning compares A5 to A3+, cut pieces and roll feed. 1/2 A4 (2 pieces per sheet) means a half-A4 measurement piece, not a split TIFF. Whole sheet uses the entire sheet. Dimensions and areas display one decimal; calculations retain precision. Check that the piece fits the printer and measurement equipment.

### Print and allow to dry

Copy the target with Copy TIFF16 for printing... if needed. Print at 100% size without automatic fit-to-page scaling. Use the documented materials and driver settings. For an unconverted base target, no additional profile conversion should change the intended RGB values. Save the print settings and follow the project drying time.

### Find the right row when measuring

Use row and position labels to orient the sheet. Labels on both sides make scanning easier. A previously exported TIFF does not change when the program code is updated; inspect the actual file you will print.

Why this step matters: incorrect scale, colour management or paper settings can produce measurements that do not describe the printer combination you intend to profile.

## ROW 3 / MEASURE CHART

# The measurement window

Measurements use a spectrometer supported by the measurement workflow. Instrument calibration and the correct target-file association are essential.

### Before starting

Choose Measure / select measurement revision. In the measurement window, check Chart (TI2), Scan mode, Scan tolerance, Instrument port and Measurement condition. The target file must match the printed sheet. Instrument port 0 means the default selection.

### Start and calibrate

Start measurement starts the instrument workflow. Calibrate when requested. At page changes, instructions show the page, row number and FORWARD or REVERSE, consistently with the row prompts. Position the correct page and use Page loaded. Instrument messages appear in the log.

### Scan and retry

Single direction, Alternate rows and Forward + reverse average describe different scanning patterns. Follow the chosen mode. Reread / retry repeats a reading. Accept reading accepts it when requested. Previous row, Next row and Next unread help you navigate. Use the sheet labels and status instructions, not only letter codes in the log.

### Finish with a saved result

Save and finish ends the measurement and saves a new revision. The overview of all patches may open after Save; this is intended review and does not start remeasurement. Earlier revisions remain available. If no successful save was recorded, no automatic import occurs; read the log before continuing.

### Measurement conditions and instrument identity

The window distinguishes conditions including M0 for i1Pro 2 without a UV filter and an unspecified instrument-default mode. Do not specify M1 or M2 unless the instrument workflow actually provides it. Model and serial number are read from instrument output when available; do not replace unknown details with assumptions.

### If the first connection fails

For some early startup failures, the measurement workflow can make limited retry attempts. This is not the same as a full hardware reset. If an error persists, cancel cleanly, read the log and check the connection and whether another program is using the instrument. Then start measurement again.

**ROWS 3-4 / SAVED REVISIONS**

# Revisions, review and remeasurement

A measurement displayed on screen is not automatically the revision used for profiling.

## Saved revisions: select the saved input

After measurement or import, the revision selector lists available saved measurements. Identify the correct session by date, target and coverage. Use revision makes the selected revision the workflow input. Check which revision is now shown as being used.

## Row 4: review and accept

Review and accept measurement opens the measurement overview and associated assessment. Review missing values, obvious deviations and patches needing attention. Enter relevant notes and finish using the explicit acceptance action. Closing the preview is not approval.

## When a patch needs remeasurement

Remeasure patch rereads one patch. Remeasure row lets you choose a page and printed row and retains the original scan mode, including forward/reverse. Review changes before Accept row replacement. Discard / Close leaves the row unreplaced. An accepted row is saved as a new revision; original and raw readings remain preserved. See page 24.

## Absolute colour and change are different

Measured Lab describes the colour itself through L\*, a\* and b\*. The change between two measurements describes how those values changed.  $\Delta E_{00}$  summarises the colour difference. Read the label before interpreting a number as a new absolute value, a component change or an overall colour difference.

## Changed measurements affect the workflow

Using a new revision can make locked inputs, the profile, checks and approval stale. This is intentional: the finished profile must be traceable to the measurement actually used. Earlier revisions provide history but do not replace updating dependent steps.

Always check three things: has the remeasurement been saved, is the new revision selected, and has the review been accepted? Refresh can show the current state but does not perform these actions.

## ROWS 5-7 / B1, B2 AND B3

# Build an ICC candidate

An ICC profile describes the relationship between printer RGB values and the colours reproduced. The candidate then needs to be checked.

### Row 5: B1. Lock profiling inputs

B1 locks the selected, reviewed measurement together with the necessary target and project information. This creates a traceable starting point so that profile building does not accidentally mix revisions. Repeat the step when the app indicates that the inputs have become stale.

### What is a RoleFile?

A role file describes how patches should be used, for example for training, control or reserved assessment. If the dialog offers No separate role file and you do not have a purpose-made role file, choose that option. Do not invent roles or select an unrelated file. For app-generated refinement targets, role associations are part of the saved evidence.

### Row 6: B2. Create recipe

Profiling recipe is used by the manual route. Measurement data smoothing controls optional pre-regularization. Smoothing at ICC build controls final colprof -r; empty means 0.5%. These stages are separate. Printing details come from Project details. Review profile name, colour data and table quality. See pages 21-23. Perceptual compression (%) controls perceptual gamut mapping; see the new rendering-intent section.

### Row 7: B3. Build ICC / automatic iteration

Manual uses the B2 recipe. Automatic creates its own recipes and does not use B2's gradient checkbox. After a successful manual build, the RGB gradient window can open if selected in B2 and pre-regularization is enabled. Saving B2 does not open it. Complete means calculation finished, not approved print quality. Automatic carries forward B2 perceptual compression; preprocessing and the gradient checkbox remain part of the Manual route.

### What is saved when it finishes?

The project contains the ICC candidate linked to locked inputs and the profiling job. This is the internal working copy. Later export creates a separate delivery copy with the chosen name and location.

### Why is creating an ICC file not enough?

A successful build shows that a profile could be calculated. It does not by itself show how a new print will turn out. Numerical checks follow, and the full verification route also requires a separate verification print and measurement.

## ROW 8 / C1. FIT, GRID AND C1

# Understand the numerical checks

These checks examine the calculated model and its tables before you assess an actual verification print.

| Item                   | What it helps you understand                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Fit                    | How closely the model agrees with the measurement data used for comparison. Training data are not an independent verification print. |
| Grid                   | Profile behaviour on a chosen calculation grid. Denser sampling can reveal detail but remains a numerical investigation.             |
| Forward and inverse    | Forward calculates colour from RGB. Inverse calculates RGB for a desired colour. Results from different engines can be compared.     |
| Local probes and ramps | How RGB or colour changes with small perturbations and along transitions. Strong changes need to be interpreted in context.          |
| Mean, median, P95, max | Average, middle value, 95th percentile and greatest deviation. Read both typical behaviour and individual extremes.                  |

## Distinguish the error measures

$\Delta E_{00}$  measures colour difference using CIEDE2000. RGB deviation in percentage points describes a difference in device values. A small difference between two calculation engines can coexist with a larger difference between desired and printed colour. These measures answer different questions.

## Is a large step a discontinuity?

A steep transition or interpolation knot is not itself proof of a discontinuity. Investigate several step sizes and neighbouring samples. Finite numerical sampling cannot prove continuity throughout the colour space.

## Next action

Open the saved check results. Read any warnings and their supporting evidence. Continue with C2 to check a print, or later use the explicitly limited numerical delivery route if that is your documented decision.

A small numerical deviation is not the same as a small print deviation. Always distinguish calculated results from what was actually measured on paper.

## ROWS 9-12 / C2, C3 AND FEEDBACK

# Check the actual print

### Row 9: save the verification target

C2 creates a verification target for the current ICC. Lab references are converted once to printer RGB with absolute colorimetry and no BPC. Preparation displays activity and elapsed time. Save TIFF16, print separately and allow it to dry. C2 contains colours, neutrals, challenges and repeats, not only problem regions.

### Avoid double profile conversion

New C2 files contain the printer profile as an embedded tag. The tag does not change RGB values or mean the profile should be applied again. Keep it without conversion. Older files may be untagged. Printer Manages Colors or a greyed-out Off control does not prove the complete print chain has no extra conversion. See page 25.

### Row 10: measure and select a revision

Measure / select verification revision uses the verification target definition. Measure, save and select the correct revision just as for the base measurement. A target from an earlier ICC candidate does not automatically verify the new candidate.

### Row 11: C3. Compare model and print

Measured vs desired compares print with desired colour. Model vs measurement compares ICC prediction with print. Both are colour differences, not absolute colour values. Desired Lab and Measured Lab contain L\*, a\*, b\*. Largest errors appear first. C3 may warn about suspected double colour management; this is a hypothesis, not proof. See page 26.

### Row 12: Review feedback

Open and read the feedback, then document your assessment using the intended completion action. A remaining Ready status may mean you only viewed the result. Go back and check that the assessment was actually saved.

### What does not approved mean?

A statement that colour management or acceptance criteria are unverified means the result does not yet have a documented approval decision. It does not by itself mean measurement failed. Read the available evidence and outstanding requirements before choosing the intended use.

A high Measured vs desired but low Model vs measurement may suggest that the model predicts the limitation reasonably well. A large model error warrants closer review. Neither case establishes the cause without further evidence.

## ROWS 13-14 / VERIFIED ROUTE

# Decision, ICC and certificate

### Row 13: Approve for intended use

Open the current check results before approving. Describe how the profile will be used and why the evidence is sufficient for that purpose. Complete the required fields and confirmation, then save the decision. Merely typing accept into text that is not saved does not complete the step.

### When approval becomes stale

Changes to the ICC, inputs or checks may mark approval as stale. Read which dependency changed and update the required steps. An old decision must not simply carry over to a new profile.

### Row 14: choose what to save

Use Save ICC and measurement certificate for delivery. The profile already exists as a working result in the project. The ICC file chooser concerns the separate copy you want to use or hand over. Choose a destination and check the filename.

### Profile names in files and image applications

The export name follows the project name and date, for example MyProject\_261003.icc. During export, the internal profile description is adapted to the delivery name so it is recognisable in applications such as Photoshop. Renaming the file later in Finder does not automatically change its internal ICC description.

### Certificate contents

The measurement certificate describes the project, materials, printing conditions, profile and check results. It should identify the iteration, the measurements supporting it and how FWA was handled. Instrument identity is reported when available. Report colour patches are screen previews, not substitutes for the original print.

### Where are reports saved?

The project retains its results. External delivery saves the report and supporting files in a common report folder. Move the entire folder for the HTML version. The PDF can be opened separately. Use the recorded path and Open report to find the project report.

The PDF certificate includes a date and space for a handwritten signature. Legal text is placed in an appendix after the signing section so that results and decisions can be read together.

## ROW 15 / ERRORS OR IMAGE

# Select new colour patches

A refinement should investigate something you want to understand or improve. It does not need to target a particular colour.

### From verification errors

From verification errors uses available verification and feedback. Describe why you want to continue. A request for a reason does not mean the app has proved another iteration would be useless; it documents your decision.

### From an image

From image extracts representative colours from an RGB image. Choose an image with an embedded ICC profile. If it has none, the app warns you and offers Use sRGB, Choose another image or Cancel. sRGB is then an explicit, recorded assumption. The original image is not changed.

### Image region and patch count

The dialog lets you limit the image region and number of new patches. A minimum RGB distance reduces close duplicates. A neighbourhood radius can add samples around selected colours; a zero radius gives image colours without these neighbours. An image containing many similar pixels does not necessarily need as many separate patches.

### Review suggestions

The selection window shows calculated measures and colour patches. The default filter is an estimated local fit deviation  $\Delta E_{00}$  greater than 5. You can change the threshold. Only visible, selected suggestions proceed. If none appear, review the threshold and chosen measure; it is not automatically proof of perfect colour.

### What does estimated error mean?

Estimated local fit uses nearby training measurements to describe local model fit. It is not a new measurement of the printed image. Model mapping  $\Delta E_{00}$  is a different measure: the deviation in the profile's calculated reproduction of the desired image colour. Missing evidence for a local estimate must not be interpreted as zero error.

### Save the selection and choose paper

Confirm the selected patches and review the suggested paper layout. Use saved image patches can reuse a saved selection. The app saves inputs and choices for traceability. Read the patch count and TIFF path in the result before printing.

## ROW 15 / ARGYLL OR INKPROF

# Choose the error-driven refinement method

## Choose the method in row 15

Choose From verification errors, then 1. Argyll only or 2. Current InkProf (Jacobian sampling). Current C3 verification and feedback are required. Argyll only is the default. This selector is for additional patches, not B3.

### Argyll only

The checked ICC preconditions Argyll target to select new device-RGB patches, without InkProf Jacobian sampling or pre-regularization. The continuation builds a high-quality colprof candidate with -r 1.0.

### Current InkProf (Jacobian sampling)

C3 residuals and Jacobian-guided patch selection are used, followed by comparison of ICC candidates using development measurements. Both methods build the ICC with Argyll colprof; they differ in patch selection and continuation settings.

### Saved choice and verification

The method is saved in proposal.json as refinementMode and passed to row 17 after the new target is printed and measured. Choose another method when creating a later error-driven refinement; existing targets and measurements are not rewritten. Image-guided refinement uses a separate path. Numerical B3 fit checks are not independent print verification: print and measure the new C2 target to assess the new result.

**ROWS 15-17 / COMBINED TARGET**

# Measure the refinement and iterate

## C2 is included until it is measured

When the current C2 verification has not yet been measured, its patches can be included with the image suggestions in the refinement target. Printing C2 is not enough to exclude it. A valid registered measurement for the current C2 determines whether it is still pending.

## Patches have different roles

The combined target preserves C2 device values and references. Image suggestions and additional control patches are added. An image  $\Delta E00$  filter does not remove pending C2 patches. The total patch count may therefore exceed the number of selected image colours.

## Row 16: Measure refinement

Print the correct TIFF16 and measure the combined target using its matching target file. Save and use the revision. When target and measurement are correctly linked, the same measurement can also provide C2 evidence and the C3 comparison for the preceding candidate. You should not need to measure the combined sheet twice.

## C2 measurements in training

C2 colour, gray and challenge patches in the combined data may be used to train the next profile. Repeats and paper-white patches are handled as controls to prevent unintended double weighting. The saved target roles determine their use.

## Row 17: Build next iteration

The next iteration uses the previous locked inputs and the new training patches specified by the saved continuation plan. The app builds a new ICC and performs associated checks. When calculation finishes, choose comparison, new physical verification or the limited delivery route on row 19.

## An important boundary between training and verification

A C2 measurement can assess the profile that created its target. Once the same measurement is used to train the next ICC, it is no longer an independent check of that new ICC. Independent checking requires new verification evidence. Do not assume separately measured C2 data are automatically included in every later training run; read the current continuation plan.

Row 17 creates the next profile inside the project. This does not automatically mean external delivery. For verified delivery, continue through current verification and rows 13-14; for a numerically limited delivery, use row 19.

ROW 18 / COMPARE WITH PREVIOUS ITERATION

# Compare two iterations

Comparison helps you see what changed and document why you continue or stop.

## The right profiles without searching filenames

The workflow uses the project's registered iterations and ICC files. Check the displayed profile names and iterations. Comparison is available when a previous and current candidate exist, normally from iteration 2.

## Different measurement meshes can be compared

The profiles do not need to have been built from identical measurement grids. The tool evaluates them at common calculation points, putting a comparable question to both models. It does not make their original physical measurements identical or equally independent.

## The 2D view starts at $L^* = 50$

The flat view shows  $a^*$  and  $b^*$  at the selected lightness  $L^*$ . Change  $L^*$  to investigate lighter and darker regions, and use the slice width to control which points appear. Turning off 2D displays a rotatable 3D view.

## Colours in the comparison

Blue represents the previous profile and orange the current profile in the iteration comparison. Always read the legend and profile names. A wider extent in a region means the models describe it differently; that alone is not a quality rating.

## Interpret differences in context

Small profile differences may support stopping when the result meets the intended use. Larger or unexpected differences may warrant further review. Compare numerical measures, local transitions and available measurements. A numerical change does not itself show whether new prints are better.

## Document the decision

Save the comparison as evidence and explain which observations mattered to your decision. When comparing error measures between iterations, identify the measurements and conditions they came from. This also makes later deterioration or divergence understandable.

A useful decision rationale is specific: what changed, what evidence was assessed and why is it sufficient for the intended use? Do not equate a larger gamut with lower measured error.

## FWA / OBA WITH ARGYLL SMOOTHING

# M0, optical brighteners and averaged white

## What is measured, and what is estimated

Native i1Pro2 M0 spectra are measured under the instrument's illumination. Argyll estimates their response under simulated D50 using paper white and instrument information. This is the best available M0-based estimate in this workflow, not direct UV excitation quantification or a native M1 measurement. A UV LED observation indicates brighteners but does not quantify the correction.

## Several white patches

All device-white spectra are averaged by Argyll (RGB channels above 99.9%). Their count, mean spectrum and sample variation are saved. Averaging can reduce random variation; it cannot restore missing UV excitation or guarantee that weak fluorescence is resolved. White controls preserved in B1 source data can supply the reference even when excluded from fitting.

## Compensation before smoothing

New B2 recipes integrate and FWA-compensate spectra once using spec2cie. Both the intermediate Argyll pre-smoothing model and the final ICC read the resulting XYZ without repeating compensation. Fit and C3 use the same policy. Save a new B2 recipe; use Manual B3 for the explicit pre-smoothing choice. Automatic iteration still creates its own recipes.

## If no white measurement is available

B2 offers Measure blank paper, Select saved blank-paper reference or Cancel. Use the same unprinted stock and backing; measure several clean locations with stationary i1Pro2 M0, then explicitly accept the mean. Follow calibration prompts. The instrument, wavelength grid and calibration standard must match. Required colour spectra must cover at or below 400 nm through at least 700 nm; measuring only white cannot repair missing colour spectra.

## What is saved and reused

The approved reference includes source readings and checksums. A separately measured white supplies an auxiliary profile white anchor; it is not presented as a target colour or independent verification patch. Analytical comparisons retain the original target patch set. B2's approved reference is available to automatic builds and continuation on the same stock when needed; current target whites take precedence.

## Certificate and final check

The certificate records white-reference source and count separately for training and verification and explains the M0 limitation. Original spectra remain unchanged. Compensation or averaging is not proof of improved reproduction: finish with a new C2 print and C3 measurement before approving the profile.

## B2 / B3 / REFINEMENT SETTINGS

# Choose the profiling and refinement path

### B2: two current choices

Argyll only (raw measurements) disables separate pre-smoothing, but colprof still fits and smooths. Argyll colprof pre-smoothing (experimental) adds an intermediate model before the final ICC build with FWA integrated once before both passes in new recipes.

### Two independent noise settings

Assumed noise (%) is first-pass colprof -r, default 0.5%. Smoothing at ICC build is final-pass -r; empty uses the documented engine default 0.5%. Neither value is an acceptance limit.

### B3 Manual versus Automatic

Manual executes the saved B2 recipe. Automatic creates its own candidate recipes with pre-smoothing off; the app carries over perceptual compression, not the whole B2 recipe. Its default Argyll mode uses high quality and final -r 1.0. Use Manual to test explicit B2 smoothing settings.

### Refinement method in row 15

Argyll only uses preconditioned target patches without Jacobian sampling; continuation builds with -r 1.0 and no pre-smoothing. Current InkProf (Jacobian sampling) uses C3 residuals and Jacobian-guided patches. Both build ICCs with Argyll. InkProf continuation disables pre-smoothing and compares final -r candidates 0.5, 1.0 and 1.5 with development data; without such data it uses the engine-default build.

### What continues into the next iteration

Row 17 follows the mode saved in the proposal and its specified combined training set, not the entire older B2 recipe. Read the selected candidate recipe and job log. Change refinement mode by creating a new proposal in row 15 and measuring its target.

### Two practical routes

First profile: rows 1-6, choose B3 Manual or Automatic, then rows 8-14 for measured delivery. Refinement: row 15 method choice, print and measure in 16, build in 17, compare in 18, then repeat 8-14 for final measured approval. Row 19 is numerical-only delivery.

### Evidence belongs to a specific profile

Check iteration, ICC and measurement revision. A changed profile cannot inherit earlier physical verification or approval. Small prediction changes or numerical stopping do not prove print accuracy. Investigate Stale/Locked status and repeat dependent checks.

## INTERPRETING FINAL RESULTS

# Best practice: acceptance and print-system limits

A useful photographic profile may still fail to reproduce some reference colours within the selected comparison limits.

### Agree on the intended use

Set the use, relevant colours, gradients, viewing conditions and acceptance criteria before judging results. Stabilise the printer and record paper, ink, preset, drying, backing and measurement conditions. Resolve acquisition errors before changing smoothing or refinement.

### Keep three kinds of evidence separate

Training fit describes agreement with profiling measurements. Profile comparison describes changes between predictions. Final C2/C3 measures a new print against desired colours and current predictions. Once a measurement trains a profile it is no longer independent verification of that profile.

### Stop for a documented reason

Use comparable errors and visual checks of neutrals, skin tones, shadows and gradients. No useful improvement may justify stopping refinement, but does not prove print accuracy or physical gamut limits. Record regularisation and the chosen Argyll only or InkProf/Jacobian refinement method.

### Interpret remaining errors honestly

Large desired-colour errors with small prediction errors may be consistent with system limits, but first check profile identity, print conversion, measurement conditions, inversion and local model reliability. Model-reachable, outside-model-or-inversion-unresolved and unknown colours are distinct. Paper white, attainable black and chroma may limit reproduction. An unresolved inverse alone is not proof of physical impossibility.

### Approval does not waive a standard requirement

Disclose unmet comparison limits. The ISO-related references shown in InkProf do not establish conformity of its custom RGB target. Where an applicable standard is required and its requirements are unmet, do not approve for that purpose. A reviewer may approve a measured photographic result for another stated use with explicit limitations.

### Present the full result

Report all assessed colours and mean, median, P95 and maximum errors, with difficult colours separately identified. Neither worst patches alone nor the overall average is sufficient. Follow the final measured approval route in rows 8-14. Row 19 supplies only numerical evidence for the current profile.

## FINAL PRINT VERIFICATION / ROWS 8-14

# Finish with a measured and approved profile

When further refinement gives no useful improvement, finish by printing and measuring a control target for the exact profile you will deliver.

### Decide when to stop

Use row 18 and comparable error measures to document why you stop. Training fit, model change and physical print accuracy are different measures. Small changes or numerical convergence alone do not prove print accuracy. This procedure does not assume automatic detection of stalled improvement.

### 1. Check and create the final target

After row 17, run row 8 (Fit, Grid and C1) for the selected current profile. Run row 9 (C2: save verification TIFF16). Check that the target belongs to this profile and iteration. The ICC transform has already been applied once.

### 2. Print without another colour conversion

Follow the accompanying target text. In Photoshop choose Discard the embedded profile (don't color manage) when opening the target. Print at 100%, with colour management disabled in the application and colour adjustment disabled in the printer driver. Keep the documented paper, ink and printer settings and allow drying.

### 3. Measure and review

Run row 10 to measure the final control print. Run row 11 (C3) to compare measured print colours with desired colours and profile predictions. Run row 12 to review feedback. This final evidence must belong to the delivered profile; measurements used to train it are not independent verification of it.

### 4. Approve and deliver

Run row 13 (Approve intended use), enter the intended use, accepted limitations and review notes, and confirm explicitly. Run row 14 to save the ICC and measurement certificate. The certificate records actual print measurement and user approval tied to that profile. Approval is for the stated use, not automatic ISO certification.

Do not use row 19 for this route: it documents a numerical-only delivery without current print verification. If the final measurement does not meet the intended use, investigate or refine before approving.

## ROW 19 / A CERTIFICATE WITH EXPLICIT SCOPE

# Finish without another test print

### When this option fits

You may be satisfied with a later iteration after numerical checks and comparison, while choosing not to print and measure another C2 target. Row 19 documents that decision and lets you save the ICC and measurement certificate with the correct scope.

### What the user does

Make sure numerical checks exist for the current iteration. Open Save ICC + measurement certificate without print verification. Enter the intended use and rationale. Confirm the scope in the dialog and choose destinations for the delivery files.

### The certificate must state this clearly

Numerically checked; this iteration has not been verified by a separate print and measurement.

### Earlier measurements retain their identity

The last measurement results may be included as background. The measured iteration and profile must be identified. Older physical results do not become new measurements merely because they appear in the current certificate. A profile comparison may explain the decision to stop without serving as physical verification.

### PDF and HTML

PDF provides a fixed document for handover and signature. HTML provides interactive profile viewing: a 2D slice initially at  $L^* = 50$  and an option to switch to 3D. PDF renders figures as still images. Move or send the entire HTML report folder so its supporting files are included.

### Signature and legal appendix

Results, scope and decision precede the signing section. Legal text follows in an appendix. It covers matters including information supplied by the client, responsibility for client-printed targets and limitations of the printer, paper and ink. Read the appendix together with the documented scope when handing over the certificate.

The document is still called a measurement certificate. Its title must not obscure the difference between measured print verification and an iteration that has only been checked numerically.

## STATUS, ACTIVITY AND RECOVERY

# When something seems to stop

| What you see                      | What to do                                                                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Ready after closing a window      | Check that you actually used Save, Use revision or the acceptance action. Opening a preview is not enough.                                   |
| Running / calculation in progress | Read the active operation and elapsed time. Let it finish. Do not open the same operation repeatedly.                                        |
| Waiting for a dialog              | Check whether a file chooser, confirmation or measurement window needs your response. Follow the status line.                                |
| Stale / Requires approve: stale   | Open the results and identify changed inputs. Repeat the necessary checks and make a current decision.                                       |
| Cannot find a new TIFF            | Read Saved results for the correct iteration and step. Use Open selected step results or Copy TIFF16 for printing....                        |
| Project integrity check failed    | Read which files are identified and open the results log. Do not attempt recovery by arbitrarily editing JSON or deleting integrity records. |
| Results exist but the list is old | Press Refresh after the current operation finishes. Then check step status and the revisions being used.                                     |

## Visible activity in later steps

Longer calculations display operation and elapsed time, including C2 colour and paper preparation. TIFF generation can show a clock and Cancel. Time is elapsed time, not a forecast. Long TIFF footer summaries are shortened when needed; full filenames and metadata remain preserved.

## The measurement window has its own signals

Calibration, row reading and instrument questions appear in the measurement status and log. The program may be waiting for a physical action. Distinguish this from profile and report calculations.

## Reporting a problem

State the step, iteration, what you clicked and the exact error text. Include the relevant part of the results log. This helps identify the cause without guessing or changing measurement data.

## REFERENCES AND TECHNICAL BASIS

# Glossary, sources and version

| Term                     | Meaning                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ICC                      | International Color Consortium; here, a profile describing colour conversion for a device or colour space.                    |
| CIELAB / L*, a*, b*      | Colour space from the Commission Internationale de l'Éclairage. L* describes lightness; a* and b* describe colour directions. |
| $\Delta E00$ / CIEDE2000 | A colour-difference measure. Always read the number together with the two colours being compared.                             |
| TIFF16 / RGB             | An image with 16 bits per colour channel; RGB denotes red, green and blue.                                                    |
| T11, T12, T13            | ArgyllCMS formats for colour patches, layout/measurement definitions and measurement data, respectively.                      |
| JSON / JSONL             | Structured text for project data. JSONL stores separate records line by line, such as an event log.                           |
| FWA / OBA / BPC          | Fluorescent Whitening Agent / Optical Brightening Agent. BPC means Black Point Compensation.                                  |
| RIP / mesh               | Raster Image Processor: software that processes print output. Mesh here means a grid of sample or calculation points.         |

## Standards as references

Project documentation relates quality assessment to ISO, the International Organization for Standardization. References to bvdM mean Bundesverband Druck und Medien, the German printing and media industry association. An internal threshold, such as  $\Delta E00$  greater than 5 in a list, must not alone be read as full ISO approval.

## Open source and licences

InkProf is distributed under the GNU General Public License version 3 or later. MATLAB and other external components have their own terms. Read LICENSE and THIRD\_PARTY\_NOTICES in the project for licensing and third-party references.

## Basis of this handbook

The Swedish and English handbooks share the same chapter order and corresponding content. The 7 October 2026 update was checked against workflow, recipe, row remeasurement, RGB gradients, C2/C3 source and usage guides. Separate diagnostic scripts are not built-in app features. Documentation and numerical tests are not new physical validation.

[Public project and installation](#)

[User documentation](#)

[InkProf licence](#)

[Third-party code and references](#)

## v1.0.0 | MATERIALS AND SHADOWS

# More samples in dark colours

### Matte paper

Project details ☐ Profiling offers Extra shadow detail for Matte. It is used when Paper finish is Matte. Standard distribution disables this adaptation.

### Editable starting values

Dark patch emphasis = 2, Shadow grid emphasis = 1.3 and Extra iteration patches = 48. The first target redistributes the chosen patch budget. The next iteration target adds up to 48 dark training patches without removing colours from the image or C3.

### Selection follows the profile black point

The current ICC profile estimates Lab. Selection covers the darkest quarter between model black and white, so it also works when matte paper has a relatively light black. RGB16 duplicates are avoided. Actual count, colours, limits and profile hash are saved in shadow-patches.json.

### Measure and compare

Denser shadow table resolution gives less emphasis to lighter regions. This is an InkProf strategy using documented Argyll tools, not an official matte-paper recipe. New patches must be printed and measured. Rebuilding a profile alone does not create new measurement data.

### Ink and coating belong to the combination

Specify dye/pigment/mixed/unknown and whether the printer applies coating: off, on, automatic or unknown. Describe the product and settings. These details follow the project and measurement certificate; the app does not change the printer.

**v1.0.0 | DELIVERY AND LICENSING**

# Gamut, certificates and source code

## Surfaces and measurement points are different views

View gamut shows an ICC-derived surface in CIELAB D50. It complements measurement/model points and the profile comparison's 2D slice, initially at  $L^*=50$ . HTML supports rotation; PDF shows a fixed figure. The surface is not a new measurement result.

## Deliver a complete report

Save ICC and the measurement certificate through row 14 after current print review, or row 19 after an explicitly limited numerical decision for a later iteration. HTML delivery is a folder with supporting files. Keep the entire folder. The signing section is followed by Appendix A with explanations/references and Appendix B with legal terms.

## Open source

InkProf code and original documentation are licensed under GNU GPL version 3 or later. Copyright 2026 Christer Törnkvist. The program is provided without warranty under LICENSE. MATLAB, Python, ArgyllCMS, fonts and other dependencies have their own terms; see `THIRD_PARTY_NOTICES.md`.

## Release status

This handbook describes release candidate 1.0.0-rc.3, dated 8 October 2026. It is not the stable v1.0.0 release. macOS is tested; Windows/Linux and every instrument mode are not qualified. External runtimes are installed separately.

## NEW FEATURES | 2026-10-07

# Two stages of regularization

## Stage 1: Measurement data smoothing

Choose Off, Argyll colprof model or an InkProf-owned model. Pre-regularization creates separate derived input before the final ICC. Original spectra, XYZ and raw measurements remain preserved. The feature is experimental and off by default.

## Argyll: Assumed noise (%)

The first colprof model uses an assumed average deviation through -r, normally 0.5%. This is not an exact measured Delta E00. The model is evaluated at the original RGB values.

## Stage 2: Smoothing at ICC build

This controls final colprof -r. Empty uses 0.5%. The value is independent of the first-stage noise assumption. Percentages display one decimal without rounding stored precision.

## FWA/OBA and B1

New B2 recipes support native M0 FWA before Argyll pre-smoothing and final fitting. Compensation is applied once, not again to derived XYZ. Changing FWA preserves B1 but requires new B2 and later results. See the M0 white-reference section.

## NEW FEATURES | 2026-10-07

# Test RGB gradients

## B2 selection and Manual B3

With pre-regularization selected, tick Open RGB gradient test after profile build. The window opens after successful Manual B3. Save recipe in B2 does not start the check. Automatic B3 creates its own recipes and does not use this B2 choice.

## Run RGB check

Nine RGB paths with 1,025 samples each are evaluated through the final ICC. Select a path and relative or perceptual intent. Curves show RGB, Lab, adjacent Delta E00 steps and Lab second differences. JSON and Markdown are saved in the job's checks folder.

## Check blue-sky inverse

The blue-sky test uses B2A and returns through A2B. It complements the forward check and changes neither recipe nor ICC. Reopen a closed window with `inkprof.checkRGBGradients(jobFolder)`.

## Interpretation

Large local steps, reversals and second differences identify places to investigate. Real printer curvature can also produce these values. No universal pass threshold is used. Printed gradients and documented viewing conditions are needed for physical assessment.

## NEW FEATURES | 2026-10-07

# Remeasure a whole printed row

## New measurement or earlier revision

Measure starts a new measurement. Saved revisions selects an earlier saved revision. Import file reads an external file. For isolated problems, use Remeasure patch or Remeasure row in the overview. A new save preserves previous history.

## Select page and row

Remeasure row uses the printed page and row number. B24 means position B, row 24; patch ID is separate. Original scan mode, tolerance and instrument conditions are retained. An isolated session measures only the selected row.

## Measure and review

Click Measure selected row, calibrate and follow FORWARD/REVERSE. Save supplies candidate values. Change Delta E00 compares previous and new measurements, not error against desired colour. Accept row replacement creates a revision replacing only this row. Discard / Close retains previous values.

## Preserved traceability

Original data, new raw readings and the decision are saved. Old paired readings are labelled historical after row replacement. Select and review the correct workflow revision before locking B1 again; Accept in the remeasurement window does not necessarily select profiling input.

## NEW FEATURES | 2026-10-07

# C2: profile, TIFF and print chain

## Conversion and tag

Lab-based targets use B2A, absolute colorimetry and BPC off. Layout and RGB16 quantization follow. The exact printer ICC is now embedded as a tag without another pixel conversion. Earlier exports do not change when code is updated.

## Keep already converted values

When opening in Photoshop, discard/ignore the embedded profile without converting RGB values; do not convert to a working space. Print at 100% without further colour conversion. Assigned sRGB or Adobe RGB followed by conversion can greatly change older untagged targets. Read PRINTING.txt and document the chain.

## Contents and difficult colours

The profiling-workflow default is 80 colours, 24 neutrals, 12 challenges and 12 repeats. Counts can be changed. Challenges may be outside gamut or have an unresolved inverse. C2 therefore does not contain only previous problem colours.

## White margin

An unprinted margin uses pure device RGB white. A white image area converted with absolute intent may receive colour to reproduce the reference white point. This is not the paper's own whiteness. A converted image margin and the target background are therefore different things.

## NEW FEATURES | 2026-10-07

# Read the C3 table and warnings

## Identity and Lab

Page and Coordinate identify position. ID is identity and Role is patch type. Desired Lab is desired colour; Measured Lab is the print. L\* is lightness, negative/positive a\* green/red and negative/positive b\* blue/yellow.

## Two error columns

Measured vs desired (dE00) compares measurement with desired colour. Model vs measurement (dE00) compares ICC prediction with measurement. Similar numbers do not mean similar colours: if both are large, measurement differs from both. Near zero in the second column means close model agreement.

## Sorting and groups

Largest errors appear first. Main statistics include unique patches, including challenges, but not extra repeats. Filter groups before drawing conclusions. Small differences between repeats or forward/reverse show repeatability, not accuracy.

## Suspected double colour management

C3 simulates extra conversion from sRGB and Adobe RGB to the printer profile and warns when a simulation explains measurements much better. This is a hypothesis; other print faults may look similar. Check assignment, conversion and the driver before retraining the model.

NEW FEATURES | 2026-10-07

# Reference colours and an existing ICC

## Fixed reference set

C2 can use a Lab table, Argyll CIE/CGATS with Lab or XYZ, or TI1 with device RGB. Use only material you are entitled to use. Names and patch codes are preserved for C3. A reference name does not imply protected vendor layout or data are included.

## Lab or RGB

Lab references are converted through the ICC and test inverse, print and measurement together. Device RGB is printed as given without ICC conversion; reference Lab is the A2B prediction. PRINTING.txt and verification.json identify the method.

## New project: Verify existing ICC

The separate seven-step workflow imports an RGB printer profile, creates a target, measures, analyses, reviews, records assessment and saves a certificate. It does not rebuild the profile. The editable 575-patch suggestion covers skin tones, shadows, neutrals and broad colours.

## Unknown independence

If original training data are unavailable, independence is unknown. Numerical checks and a good photographic impression do not give automatic approval or ISO certification. Document intended use, the chain and limitations.

## Methods in RGB Target

The RGB target designer offers OFPS, incremental far point, quasi-random, body-centred cubic and random through Argyll. An existing ICC can guide placement through pre-conditioning; it does not create new measurements or apply the profile to the raw target RGB. Method, seed and profile reference are saved.

## NEW FEATURES | 2026-10-07

# Diagnose with JSON and the right check

## Traceable chain

B1 locks measurement input. The job saves recipe.json, status.json and result/profile.icc. C2 saves an exact ICC copy, desired Lab, RGB16 and placement in verification.json. C3 links measurement JSON and its TI3 revision through hashes. Coordinate and ID are different identifiers.

## Built-in MATLAB checks

inkprof.checkProfileFit checks the forward model against training. inkprof.checkProfileGrid and inkprof.checkProfileC1 provide grid, CMM and inverse diagnostics. inkprof.checkVerificationTarget compares the verification print. inkprof.checkRGBGradients opens the gradient window. Read each check's purpose.

## Separate scripts are not app buttons

The development measurement-JSON inverse script compared Lab, inverse RGB and return through A2B. It is not a built-in InkProf function. Small Lab-RGB-Lab error shows internal consistency, not a correct physical print. Alternative solutions can give large RGB differences.

## Test the hypothesis first

First check suspected wrong ICC or inverse numerically and through hashes, without another print. Check a suspected wrong row against TI2 and by remeasurement. A changed print chain may require a new sample with identical device RGB. Do not rebuild the model to conceal an unresolved chain error.

## NEW FEATURES | 2026-10-07

# Compare accuracy and gradients

## Open the comparison

After valid B1, choose Compare accuracy and gradients in the main window. Enter one to six positive final colprof -r values. Compare the currently offered Argyll candidates. Several builds may take minutes; progress is displayed.

## Read the results

Compare fit against original measurements, held-out evidence where available, model change and photographic-gradient diagnostics. Training fit is not independent validation. Inspect matching gradient paths: an average across different interior coverage can mislead.

## Automatic B3 selection

With development controls, Automatic tests final -r values 0.5, 1 and 1.5 at high table quality. Selection balances accuracy and float-gradient smoothness. Guards check patch and gray regressions, curvature, reversals, coverage and colour span. Defaults permit 0.05 RMS accuracy tradeoff for at least 5% gradient improvement. These are numerical selection rules, not print acceptance criteria. Without development controls, one candidate is built with no claim of superiority.

## Choose deliberately

Comparison preserves B1 and leaves the current workflow ICC selected. To use a comparison setting, enter it in B2 and build Manual B3. New recipes support FWA before pre-smoothing with the measured-white requirements described in the M0 section.

NEW FEATURES | 2026-10-07

## Photographic gradient conversion

### Open the diagnostic

Choose Photographic gradients in the main window after a successful profile build, or in the RGB gradient window. The check uses the exact selected ICC and saves JSON and Markdown in its checks folder.

### What is tested

Nine paths with 1,025 samples are converted from sRGB and Adobe RGB (1998) to printer RGB, then through relative A2B to predicted Lab. Relative and perceptual intent, BPC on/off, and float, 16-bit and 8-bit input/output buffers give 216 combinations. LittleCMS performs these conversions; Photoshop and printer drivers may behave differently.

### Read the columns

Curvature P95 measures normalized Lab second differences. L\* reversals count decreasing lightness steps on eligible interior points; interpret them in relation to the chosen path. Unique RGB codes and unchanged steps describe quantization plateaus. The JSON also records longest plateau and Lab span. Large numbers are not automatically visible defects.

### Limits of the test

Hard device clipping is excluded from comparison metrics, but this is not a proven physical-gamut mask. Quantization plateaus are expected, especially at 8 bit. Compare matching paths and precision. Numerical smoothness does not certify absence of banding; verify printed gradients.

## NEW FEATURES | 2026-10-07

# Certificate evidence

## Pre-regularization in the certificate

The certificate records the selected job recipe, pre-regularization method, its assumed noise (avgdev) and final colprof -r. Saved results include the model change when available. Raw measurements are preserved. Model change is not print accuracy. The former InkProf grid regularization and its measurement weights have been removed.

## Gradient evidence and regeneration

A saved photographic-gradient report is included only if its SHA-256 matches the certificate ICC. The certificate summarizes CMM, settings, sample count, interior coverage, reversals and curvature, and embeds full per-path evidence in JSON. Missing evidence is stated explicitly. Re-run step 14 for a new certificate; historical and signed certificates are preserved.

## NEW FEATURES | 2026-10-07

# One ICC, three rendering intents

## New B2 recipes

Newly built profiles support perceptual, relative colorimetric and absolute colorimetric in one ICC file. B2 provides Perceptual compression (%), passed to colprof -s. The default 20% generic compression is a starting setting, not a measured optimum. Automatic B3 carries forward this B2 value.

## What the ICC contains

Perceptual uses a gamut-mapped B2A0 table. Relative uses B2A1. Absolute uses the relative table plus the media white point (wtpt); it does not require a separate B2A3 table. The colour-management application selects the rendering intent. The profile header default does not prevent choosing another intent.

## Build validation and certificate

New builds require perceptual and relative tables with distinct data and a valid media-white tag before publication. Saved jobs and certificates record the rendering-intent settings. Structural validation and numerical conversions do not establish print quality.

## Update an existing profile

Restart InkProf, save a new B2 recipe and rebuild B3 to obtain the new gamut mapping. Existing recipes, ICC files and imported profiles are preserved. Recheck the profile and its print before approval and delivery. C2 verification still applies the ICC once, absolute without BPC, followed by printing with colour management disabled.

[ArgyllCMS colprof](#)

[ICC specification](#)

**NEW FEATURES | 2026-10-07**

# TIFF: conversion versus embedded ICC

## Read the companion text file

PRINTING.txt beside modern TIFF targets, and \*-README.txt for the legacy exporter, list ICC conversion and Embedded ICC separately. Tagged files identify the printer profile and its SHA-256. C2 also has package-level PRINTING.txt; read it together with print/PRINTING.txt.

## Conversion: what changed the pixels?

Base and refinement targets render source device RGB directly without a printer ICC conversion by InkProf. Ordinary C2 colour-reference targets have already passed through the printer ICC once, absolute colorimetric without BPC. C2 device-RGB reference sets instead print their RGB unchanged; the ICC predicts reference Lab only. Imported source history may be unknown.

## Tag: what identifies the file?

Both kinds of C2 TIFF embed the printer ICC as an identifying tag. Embedding alone changes no pixels and is not an instruction to apply the ICC again. A tag does not prove that no other conversion occurred earlier. InkProf records its own processing separately. Base and refinement targets normally have no embedded profile.

## Print without another conversion

Print at 100% with colour management disabled in the application, OS print path and driver. Preserve any embedded tag without conversion. Do not assign another profile or convert the TIFF to a working space. A program can use the embedded tag if colour management is enabled. New packages contain the clearer instructions; existing TIFFs and packages are preserved.

**COMPLETE PROFILING APPENDIX | 2026-10-08**

# Appendix: plan and create the base target

## 1. Define success before printing

Agree on photographic use or a specific reference condition, viewing light, important colours and gradients, and acceptance criteria. Record printer, ink, paper, driver preset, quality, resolution, backing and instrument. Check nozzles and allow consistent drying time. A custom RGB target does not establish ISO proofing conformity.

## 2. Use an existing ICC as a starting estimate

Choose a reliable RGB printer ICC for the same printer, ink, paper and settings, or a reasonably similar system. In target generation, select an Argyll method supporting perceptual placement and supply the preconditioning profile. Argyll uses its predictions to distribute device-RGB patches; the base target is not converted through that ICC. A poor or unrelated profile can bias sampling. Without a suitable ICC, use an unconditioned initial target and improve the sampling after the first build.

## 3. Choose the measurement budget

A practical starting recommendation is about 1,000 colour samples with broad RGB coverage, neutrals, shadows, skin tones and distributed controls. For a careful full profile, 1,500–2,000 samples are reasonable; the existing 2,000-patch route is appropriate when measurement time permits. A 300–600-sample pilot can reveal setup problems before a larger run. These are workflow recommendations, not a universal optimum or a software default. Reserve space for repeats and several unprinted-white readings; confirm the actual generated count, including controls, in the preview.

## Decision before saving

Preserve patch size and instrument scan geometry. Choose paper dimensions and inspect the actual page preview. Save the TIFF16 together with its matching TI2, layout and PRINTING.txt. Keep the original ICC and its identity with the project. A target-generation preconditioner is distinct from the profile later built from the measurements.

## COMPLETE PROFILING APPENDIX | 2026-10-08

# Appendix: print, measure and decide on OBA

## 4. Print the prepared RGB target

Follow its PRINTING.txt. In the Photoshop profile-mismatch dialog choose Discard the embedded profile (don't color manage). Disable colour management in the printing application and colour adjustment in the driver. Print at 100% with the recorded preset. Do not convert to Adobe RGB or apply the printer ICC again. An identifying embedded ICC does not change this instruction. C2 reference targets have already received their specified preparation; their RGB values must also reach the printer unchanged.

## 5. Measure and review before B1

Calibrate the instrument, use consistent backing and scan direction, and measure all pages. Save each measurement round as a revision. Review identities, white controls, duplicate colours and unusual readings; remeasure suspect patches or a whole row with the correct page and forward/reverse choice. Correct acquisition errors before smoothing. Select and accept the intended revision, then lock B1. Keep different print rounds identifiable.

## 6. Decide whether FWA/OBA compensation is needed

Do not enable compensation solely because white looks yellow or a UV lamp makes the paper glow. Consider the paper, verified measurement condition and intended viewing light. For the supported i1Pro2 M0 route, compensation is a D50 simulation using available visible spectra, not a measurement of UV excitation or an M1 result. If the paper has little OBA, or the available evidence does not support compensation, retain the uncompensated route and document the choice.

### If compensation is selected

Use native M0 spectra with verified instrument metadata and sufficient spectral coverage. InkProf averages available white-patch spectra and records their count and variation. If no usable white reference exists, B2 offers blank-paper measurement, a saved approved reference, or Cancel. Measure several clean locations on the same paper with the same backing. Averaging can reduce random noise; it cannot recover missing excitation information. Save a new B2 recipe. Keep compensation and viewing assumptions consistent through comparisons and final verification; changing this choice invalidates later results.

## COMPLETE PROFILING APPENDIX | 2026-10-08

# Appendix: build, refine and stop

## 7. Choose smoothing and the B3 route

Start with Argyll only (raw measurements) when repeatability is good. If noisy measurements or gradients justify it, compare Argyll colprof pre-smoothing (experimental) against the raw route. Its Assumed noise (%) and Smoothing at ICC build are separate settings; 0.5% is the initial recipe value, not an established optimum for every instrument. Check colour fit and equivalent photographic gradient paths. Select the RGB gradient test when useful. For an exact saved B2 recipe, choose Manual in B3; Automatic creates its own recipes and does not inherit every B2 smoothing setting. With FWA and pre-smoothing, use the newly saved supported two-pass recipe.

## 8. Select the refinement method

Use Argyll only as the initial refinement choice: Argyll places additional patches using the current profile. Current InkProf (Jacobian sampling) uses measured local residuals and local response to propose additional RGB samples. Both routes build the resulting ICC with Argyll. Choose Jacobian sampling when a reproducible local discrepancy warrants targeted investigation; do not expect it to make unattainable colours printable. Save the choice with the proposal. Print and measure the added device-RGB target unchanged, then build the next iteration.

## 9. Compare equivalent evidence

Use row 18 for Previous profile prediction vs current profile prediction at common samples. Separately review fitting error, development/held-out evidence where available, repeatability, gradients and Measured print vs desired colour. A smaller training error alone is insufficient. A small inverse round-trip error only shows consistency of the model and inverse. Large measured discrepancies require checks of profile identity, RGB values, print management, patch mapping and measurement conditions before attributing them to gamut.

## 10. Decide whether another iteration is worthwhile

Continue when a specific measured uncertainty can be reduced. Stop when comparable errors and gradients show no useful improvement beyond repeatability, or a further round produces an unacceptable accuracy/smoothness trade-off. One or two comparable rounds without meaningful gain are a practical review trigger, not an automatic InkProf acceptance rule. Record the evidence, selected profile and reason for stopping. Numerical convergence does not prove physical colour accuracy or compliance.

**COMPLETE PROFILING APPENDIX | 2026-10-08**

# Appendix: final verification and certificate

## 11. Verify the exact profile to be delivered

After choosing the final iteration, run row 8 (Fit, Grid and C1) and row 9 (C2: save verification TIFF16) for that current profile. Make a new control print with the documented settings and no further colour conversion. Measure it in row 10, retain the revision, and run row 11 (C3) and row 12 (feedback). Do this even if development or training measurements already look good. A measurement used to train a later build cannot independently verify that later ICC.

## 12. Assess the whole result

Review count, mean, median, P95 and maximum for all assessed unique colours, not only the worst patches. Identify challenge colours, controls and repeatability. Compare desired-colour error with current-profile prediction error where available. Inspect important photographs and gradients. Model-reachable, outside-model/inversion-unresolved and unclassified references describe model evidence; they are not proof of a physical gamut limit. Confirm white, black, neutral balance and the intended viewing conditions.

## 13. Approve or decline the intended use

In row 13, enter review notes, scope, print settings, limitations and the decision. Approval is justified only by the relevant measured evidence and agreed criteria. If a required standard limit is not met, report it; printer limitations do not waive that requirement. Photographic use may be approved with explicit limitations when agreed, while conformity for a different use is declined. If the remaining cause is unresolved, say so. Row 14 exports the selected ICC and English measurement certificate.

## 14. Preserve the evidence and status

Check that the certificate identifies this ICC, iteration, current verification revision, regularisation, FWA reference and reviewer decision. Retain the target, TI2/layout, PRINTING.txt, raw measurements, recipe, ICC and complete reports. Row 19 is numerical-only delivery and cannot replace fresh measured verification. If the ICC or print setup changes after approval, repeat final verification. The defensible conclusion is measured and approved for the stated use, with disclosed limits—not automatic ISO approval.

# Appendix: evidence behind the recommendations

## Argyll: sampling and measurement quality

The target documentation supports using a prior or similar ICC to guide perceptual patch placement. printtarg explains instrument-dependent patch geometry and randomised locations, which help identification and distribute spatial printing effects. Preserve the generated layout; do not rearrange patches manually. There is no universally optimal patch count. The suggested budgets in this appendix are InkProf working recommendations to be evaluated against repeatability and independent results.

## Argyll: smoothing is an uncertainty assumption

colprof defines -r as the assumed average deviation of device and instrument readings, with a default of 0.5%. Raising it gives less weight to individual readings and more to collective behaviour. A percentage is not  $\Delta E_{00}$ . Estimate the need from repeat readings and compare accuracy and gradients; do not increase smoothing simply to conceal a systematic print error.

## X-Rite: distinguish M0 compensation from M1

X-Rite describes M1 as the preferred condition for OBA-enhanced substrates and explains that M0 does not fully define UV content. The i1Pro2 hardware supports M1, but the InkProf path described here uses verified M0 data. Recording spectral bands near or below 400 nm does not measure the illumination spectrum or establish calibrated UV excitation. Averaging white readings improves repeatability, not the physical definition of M0. Use actual M1 measurements in a compatible workflow when required by the reference or conformity claim.

## Fogra: verification needs the relevant reference and control

Fogra links contract-proof verification to its defined MediaWedge, printing condition and applicable tolerances. An InkProf custom RGB verification and reviewer approval are useful evidence for a stated use, but do not substitute for that complete procedure. The independent final control print in this appendix is an InkProf quality-control recommendation.

## What external sources do not establish

These sources do not demonstrate that InkProf Jacobian sampling outperforms Argyll only, nor that the recommended patch budgets or stopping triggers are universally optimal. Those decisions require comparable local results. Sources reviewed on 8 October 2026; the handbook describes current InkProf behaviour separately from external guidance.

Argyll: [target placement and preconditioning](#)

Argyll: [patch layout and randomisation](#)

Argyll: [smoothing and profile building](#)

X-Rite: [measurement conditions and optical brighteners](#)

Fogra: [MediaWedge and contract-proof verification](#)

## COMPLETE PROFILING APPENDIX | COMPARISON

# Appendix: Torger and i1Profiler

### **Torger: a useful photographic starting point**

Torger recommends preconditioning, optimised placement and extra greys. His Colormunki example favours about 840 patches as a practical balance; this is experience with that instrument and print system, not a universal limit. He uses colprof -r1.0 to trade a little fit accuracy for smoothness and evaluates diagnostic gradients and a familiar photograph. His reported profile-matching errors compare predictions with measured device patches, including training-data checks; they are not interchangeable with InkProf C3 errors against desired colours on a separate profile-converted print. His tutorial is primarily a base-profiling workflow, not evidence that InkProf refinement replicates a particular iterative algorithm.

### **i1Profiler: a more extensive measurement route**

X-Rite's recommended RGB workflow proposes 2,033 or 1,586 patches, wider patches for better sampling, about an hour of drying, instrument calibration and saved measurement data. These recommendations concern its own software and layouts: do not blindly increase InkProf patch width or assume one hour guarantees stability for every paper. i1Profiler also offers targeted optimisation and optical-brightener compensation; the internal engine is not Argyll, and its OBC must not be described as identical to InkProf's M0 method.

### **Reconciled recommendation for InkProf**

For a paper used regularly and a careful i1Pro2 profiling job, begin with approximately 1,500–2,000 patches when feasible. Around 800–1,000 is a reasonable lower-budget start; a 300–600-patch pilot is for setup checks. Use a suitable existing ICC for placement, distribute neutral and white controls, and preserve reliable patch geometry. Compare raw Argyll fitting with justified smoothing on the same data. Choose further patches from measured evidence rather than a fixed obligation to iterate.

### **The final decision remains measured**

Neither an attractive photograph nor low model-fit error alone establishes measured accuracy for the intended reference. Keep the familiar photographic check and gradient review, then make the independent final C2 print and C3 measurement before reviewer approval and certificate delivery. This combines practical photographic assessment with traceable measurement.

[Torger: Inkjet printer profiling with Argyll and Colormunki](#)

[X-Rite: Recommended RGB Printer Profiling With i1Profiler](#)

[X-Rite: i1Profiler features and targeted optimisation](#)

**COMPLETE PROFILING APPENDIX | VISUAL ASSESSMENT**

# Keith Cooper: test images and sky gradients

## Keith Cooper: photographic test images

Keith Cooper (Northlight Images) provides printer test images and assessment notes, including the Datacolor image made available courtesy of Datacolor. Use it for overall photographic assessment of skin tones, neutrals, shadows and saturated colours. In InkProf comparisons, inspect the sky gradient for banding or uneven transitions when assessing smoothing; it is a visual diagnostic, not proof of a particular cause. Compare the same source, ICC path, intent, BPC, print preset and viewing light. The original Adobe RGB photograph needs one intended printer-profile conversion; an already converted device-RGB version must receive no second conversion. Do not apply fitting-target opening instructions to the original photograph. JPEG quantisation and clipping can also produce visible steps; use synthetic high-precision gradients alongside it. Download from Keith Cooper's page; the image is not distributed in the InkProf repository.

[Keith Cooper — Printer Test Images, Northlight Images](#)